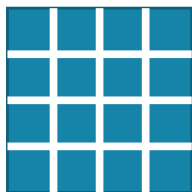


# LoRA as a low-rank matrix update

A frozen weight matrix plus a compact trainable correction

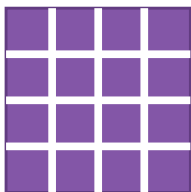
## 01 WEIGHT UPDATE

**Frozen weights**  
 $W_0 \in \mathbb{R}^{d \times k}$



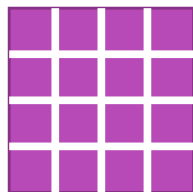
+

**LoRA update**  
 $\Delta W = (\alpha/r)BA$



=

**Adapted weights**  
 $W^* = W_0 + \Delta W$



*The backbone stays frozen; training learns only the compact correction  $\Delta W$ .*

## 02 LOW-RANK DECOMPOSITION

**B**  
 $d \times r$



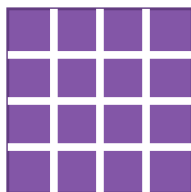
×

**A**  
 $r \times k$



=

**BA**  
 $d \times k$



$$r \ll \min(d, k) \implies \text{rank}(BA) \leq r$$

## 03 FORWARD PASS

$$x \in \mathbb{R}^k$$

**Frozen path**  
 $\text{dequant}(W_0)x$

**Trainable path**  
 $x \rightarrow A \rightarrow B \rightarrow (\alpha/r)BAx$

+

$$h = \text{dequant}(W_0)x + \frac{\alpha}{r}BAx$$

*Visual dimensions are schematic; the equations state the exact tensor shapes.*